

Name : Aarya Nilesh Patil

Batch : T21

Roll No : 87

EXPERIMENT 3

Aim : To Perform various GIT operations on local and Remote repositories using GIT Cheat-Sheet

Pull and Push Processes

The pull command used to fetch the repository from github to local while push is used to commit files from local repository to Github.

Push ☐ Push changes to Web

repository Pull ☐ Pull changes to

Local repository

The following commands are used for pull and push repositories

A) Push command

\$ git remote add origin <https://github.com/bhushanjadhav1/siesworkshop.git>

\$ git remote show origin

```
● (base) aaryapatil@AARYAs-MacBook-Air ~ % git init
  Initialized empty Git repository in /Users/aaryapatil/.git/
● (base) aaryapatil@AARYAs-MacBook-Air ~ % git remote add origin https://github.com/AaryaPatil-24/SEPM_T21_87.git
● (base) aaryapatil@AARYAs-MacBook-Air ~ % git remote show origin
* remote origin
  Fetch URL: https://github.com/AaryaPatil-24/SEPM_T21_87.git
  Push URL: https://github.com/AaryaPatil-24/SEPM_T21_87.git
  HEAD branch: main
  Remote branch:
    main new (next fetch will store in remotes/origin)
```

If you add remote again then will show you fatal error.

```
$ git remote add origin https://github.com/bhushanjadhavl/Myrepository.git fatal: remote origin already exists.
```

So, to delete origin rm origin command is used

```
$ git remote rm origin
```

Now, to push the local repository to remote github following command is used

```
$ git push -u origin master
```

```
• (base) aaryapatil@AARYAs-MacBook-Air sepm % git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 10 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 746 bytes | 746.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AaryaPatil-24/SEPM_T21_87.git
619c2df..a033fd4 main -> main
branch 'main' set up to track 'origin/main'.
```

Now you can check the github for updated contents.

The screenshot shows the GitHub interface for a repository named 'SEPM_T21_87' owned by 'AaryaPatil-24'. The repository is public and has 1 branch (main) and 0 tags. The main branch is selected, showing a commit history with 4 commits. The latest commit is by 'AARYA PATIL and AARYA PATIL' merging branch 'main' of 'https://github.com/AaryaPatil-24/SEPM_T21_87' with commit 'a033fd4' 2 minutes ago. The commit message is 'Rename Screenshot 2026-02-24 142406.png to GANTT'. The files listed are 'Gantt' (renamed) and 'index.html' (initial commit). The README section is empty, with a prompt to 'Add a README'. The right sidebar shows repository statistics: 0 stars, 0 watching, 0 forks, 0 releases, and 0 packages. The contributors section shows 1 contributor.

B) Pull Changes

Pull command is used to download the remote updated repository into local one. The command for download is

```
$ git pull
```

Now you can see the changes in

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git pull
Already up to date.
```

local repository using git log.

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git log --oneline origin/main
a033fd4 (HEAD -> main, origin/main, origin/HEAD) Merge branch 'main' of https://github.com/AaryaPatil-24/SEPM_T21_87
e4f822c Initial commit
619c2df Rename Screenshot 2026-02-24 142406.png to GANTT
4283510 Add files via upload
```

C) Fetch

Suppose you have a file in github and you have changes that.

ORIGINAL FILE

```
Code Blame 11 lines (11 loc) · 276 Bytes

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>SEPM LAB First</title>
7  </head>
8  <body>
9      <p>This is Thadomal Shahani Engineering College, Bandra West.</p>
10 </body>
11 </html>
```

Changed File

Now we use fetch command to fetch the changes, which will show you both the files like original and changed in local repository.

\$ git fetch

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 372 bytes | 372.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/AaryaPatil-24/SEPM_T21_87.git
   a033fd4..22e4915  main -> main
```

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git fetch
```

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git log --oneline origin/main
```

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git log --oneline origin/main
22e4915 (HEAD -> main, origin/main, origin/HEAD) Updated index.html
a033fd4 Merge branch 'main' of https://github.com/AaryaPatil-24/SEPM_T21_87
e4f822c Initial commit
619c2df Rename Screenshot 2026-02-24 142406.png to GANTT
4283510 Add files via upload
```

Here fetch will not show you like updated changes file as like push. So use merge command to merge the changes so use following command for merge.

\$ git merge origin/master

```
● (base) aaryapatil@AARYAs-MacBook-Air sepm % git merge
Already up to date.
```